

CliniScan: Chest X-ray Lung Abnormality Detection System

Milestone 2: Model Development & Baseline Training

Adithya Jayaram

Batch B13

Table of Contents

Section	Page
1. Project Overview & Objective	2
2. Data Preparation Pipeline	2
2.1. Image Preprocessing	2
2.2. YOLO Annotation Conversion	2
3. Classification Model (ResNet18)	3
4. Object Detection Model (YOLOv8n)	3
5. Challenges & Observations	4
6. Conclusion	4

1. Project Overview & Objective

The goal of Milestone 2 is to develop and train deep learning models capable of analyzing chest X-ray images. The system is designed as a two-stage pipeline consisting of a binary classifier and an object detector. The classifier determines whether an image is normal or abnormal, while the detector identifies the locations of abnormalities using bounding boxes.

Pipeline Overview

1. **Binary Classification Model** – Predicts whether an X-ray contains abnormalities.
2. **Object Detection Model** – If abnormal, detects the location of pathological findings.

2. Data Preparation Pipeline

2.1. Image Preprocessing

Chest X-ray images were converted from DICOM format to PNG and resized to $224 \times$ resolution for classification. Pixel intensities were normalized and the dataset was organized into training and validation folders.

Folder Structure:

- classification_data/train: normal/abnormal
- classification_data/val: normal/abnormal

2.2. YOLO Annotation Conversion

Bounding box annotations from the original CSV dataset were converted into YOLO format. Coordinates were normalized between 0 and 1 using the YOLO annotation convention:

class_id center_x center_y width height.

3. Classification Model (ResNet18)

Component	Specification
Base Model	ResNet18 (ImageNet pretrained)
Input Size	224 × RGB
Output Classes	2 (Normal, Abnormal)
Loss Function	CrossEntropyLoss
Optimizer	Adam
Learning Rate	0.0003
Batch Size	32
Epochs	20

Performance:

The classifier achieved strong validation accuracy (~95%) demonstrating effective separation between normal and abnormal chest X-ray images.

4. Object Detection Model (YOLOv8n)

Component	Specification
Model	YOLOv8n
Pretrained Weights	COCO

Image Size	640 ×
Epochs	30
Batch Size	8
Device	NVIDIA GTX 1650 GPU

The YOLOv8n model was trained to detect abnormal regions within chest X-ray images. Training produced two weight files: best.pt (best performing model) and last.pt (final epoch model).

5. Challenges & Observations

Major challenges encountered during training included:

- **Class Imbalance:** Disparity between the number of normal and abnormal samples in the dataset.
- **Small Lesions:** Difficulty in accurately localizing very small pathological findings.
- **Hardware Constraints:** GPU memory limitations during the training of the object detection model.

Despite these constraints, the models successfully learned meaningful patterns from the data.

6. Conclusion

Milestone 2 successfully implemented a working AI pipeline for chest X-ray analysis. The ResNet18 classifier identifies abnormal scans while the YOLOv8n detector localizes abnormal regions. These models provide the foundation for further improvements and refinement in Milestone 3.